

SOLVING QUADRATICS — Cheatsheet

Algebra 1 · Topics 10–13 · Esme · pre-approved by Mr. Mistry

★ WHICH METHOD? — DECISION TREE

Looks like	Use
$(\text{stuff})^2 = N$	Square Root method
Already factored / easy to factor	Factoring (ZPP)
$x^2 + bx + c = 0$, b even	Complete the Square or QF
Anything else	Quadratic Formula (always works)

When in doubt → Quadratic Formula. Universal hammer.

★ QUADRATIC FORMULA (Topic 13)

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Works for ANY quadratic. Sing to "Pop Goes the Weasel."

- Get into standard form ($= 0$).
- Identify a, b, c (**with signs!**).
- Compute discriminant: $b^2 - 4ac$.
- Take $\sqrt{(\text{discriminant})}$.
- Plug into formula: $x = (-b \pm \sqrt{\text{disc}}) / 2a$.
- Split into 2 answers (+ branch, - branch).

$$\begin{aligned} x^2 + 7x + 10 &= 0 \\ a=1, b=7, c=10 \\ \text{disc} &= 49 - 40 = 9; \sqrt{9} = 3 \\ x &= (-7 \pm 3)/2 \rightarrow x = -2 \text{ or } x = -5 \end{aligned}$$

PARENS protect: $(-5)^2 = 25$, NOT $-5^2 = -25$.

DISCRIMINANT — $b^2 - 4ac$

Tells you how many real solutions there are.

Discriminant	Solutions	Graph
> 0	2 real	crosses x-axis twice
$= 0$	1 real (repeated)	touches x-axis once
< 0	0 real (none)	doesn't touch x-axis

SOLVING BY FACTORING (Topic 10)

Zero Product Property: if $ab = 0$, then $a=0$ or $b=0$.

- Move EVERYTHING to one side; other = 0.
- Factor (GCF first, then trinomial / DoS).
- Set EACH factor = 0.
- Solve each mini-equation.
- Write BOTH solutions.
- Check by plugging back in.

$$\begin{aligned} x^2 + 5x - 14 &= 0 \\ (x+7)(x-2) &= 0 \\ x+7 &= 0 \rightarrow x = -7 \\ x-2 &= 0 \rightarrow x = 2 \end{aligned}$$

Equation MUST = 0 before factoring to solve.

SQUARE ROOT METHOD (Topic 11)

When the squared part is already isolated.

- Isolate the $(\text{stuff})^2$ on one side.
- Take $\sqrt{}$ of both sides. **ALWAYS $\pm$!**
- Solve for x.
- Simplify if needed.

$$\begin{aligned} (x - 3)^2 &= 16 \\ x - 3 &= \pm 4 \\ x &= 3 + 4 = 7 \text{ OR } x = 3 - 4 = -1 \end{aligned}$$

Two answers. NEVER forget $\pm$.

COMPLETING THE SQUARE (Topic 12)

Force $x^2 + bx + ?$ to factor as a perfect square. HSAB.

- Move c to right side: $x^2 + bx = -c$
- Compute $(b/2)^2$ — half of b, squared.
- ADD $(b/2)^2$ to BOTH sides.
- Factor left as $(x + b/2)^2$.
- Take $\sqrt{}$ of both sides (with $\pm$).
- Solve for x.

$$\begin{aligned} x^2 + 6x - 7 &= 0 \\ x^2 + 6x &= 7 \\ (b/2)^2 &= 9; \text{ add: } x^2 + 6x + 9 = 16 \\ (x+3)^2 &= 16 \\ x+3 &= \pm 4 \rightarrow x = 1 \text{ or } x = -7 \end{aligned}$$

HSAB: Half, Square, Add to Both.

Perfect Squares Lookup

n	1	2	3	4	5	6	7	8	9
n^2	1	4	9	16	25	36	49	64	81
n	11	12	13	14	15	16	17	18	19
n^2	121	144	169	196	225	256	289	324	361

Simplify Radicals (when disc isn't perfect)

Pull out the biggest perfect-square factor.

- Find biggest perfect square that divides the #.
- Split: $\sqrt{(\text{perfect} \cdot \text{rest})} = \sqrt{\text{perfect}} \cdot \sqrt{\text{rest}}$.

$$\begin{aligned} \sqrt{72} &= \sqrt{(36 \cdot 2)} = 6\sqrt{2} \\ \sqrt{50} &= \sqrt{(25 \cdot 2)} = 5\sqrt{2} \\ \sqrt{48} &= \sqrt{(16 \cdot 3)} = 4\sqrt{3} \\ \sqrt{20} &= \sqrt{(4 \cdot 5)} = 2\sqrt{5} \end{aligned}$$

Reducing Fractions in QF Answers

If both terms in numerator share a factor with 2a, simplify.

$$\begin{aligned} x &= (4 \pm 6)/2 \rightarrow x = (4+6)/2=5, \\ &\quad (4-6)/2=-1 \\ x &= (6 \pm 2\sqrt{3})/4 \rightarrow x = (3 \pm \sqrt{3})/2 \\ &\quad (\div 2) \end{aligned}$$

★ FORMULAS AT A GLANCE

Quad Formula	$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$
Discriminant	$b^2 - 4ac$
AOS (standard)	$x = -b / (2a)$
Vertex form	$y = a(x-h)^2 + k \rightarrow (h, k)$
Difference of sq	$a^2 - b^2 = (a+b)(a-b)$
$(a \pm b)^2$	$a^2 \pm 2ab + b^2$
Zero Product	$ab = 0 \rightarrow a=0 \text{ or } b=0$
$\sqrt{}$ method	$(\text{stuff})^2 = N \rightarrow \text{stuff} = \pm\sqrt{N}$
HSAB	$(b/2)^2$ added to both sides

⚠ TOP SOLVING TRAPS

- Forgetting $\pm$** when taking $\sqrt{}$. Always two answers.
- Solving without = 0.** Move everything to one side FIRST before factoring.
- Sign of b in QF:** if $b = -5$, write $(-5)^2 = 25$, NOT $-5^2 = -25$. Use parens.
- HSAB skipping "both sides":** always add $(b/2)^2$ to BOTH sides.
- Stopping at 1 answer.** Quadratics have 2 (unless disc = 0).

Vocab

Solution / root / zero / x-int	All mean "where $y = 0$ "
Discriminant	$b^2 - 4ac$ (part under $\sqrt{}$)
$\pm$ (plus or minus)	Two answers
Standard form	$ax^2 + bx + c = 0$
Zero Product Prop.	$ab=0 \rightarrow a=0 \text{ or } b=0$
Perfect square trinomial	factors as $()^2$

Sanity Check Your Answer

- Plug answer back into ORIGINAL equation.
- Both sides should match (or = 0).
- If not → arithmetic mistake. Check signs first.